

```
[1]: num1 = int(input("Enter first number: "))  
      num2 = int(input("Enter second number: "))
```

```
      sum = num1 + num2
```

```
      print("Sum =", sum)
```

```
Enter first number: 1
```

```
Enter second number: 2
```

```
Sum = 3
```

```
[2]: a = int(input("Enter first number: "))  
      b = int(input("Enter second number: "))
```

```
      temp = a
```

```
      a = b
```

```
      b = temp
```

```
      print("After swapping:")
```

```
      print("a =", a)
```

```
      print("b =", b)
```

```
Enter first number: 2
```

```
Enter second number: 6
```

```
After swapping:
```

```
a = 6
```

```
b = 2
```

```
[3]: ##### n = int(input("Enter a number: "))  
fact = 1  
  
for i in range(1, n + 1):  
    fact = fact * i  
  
print("Factorial =", fact)
```

Enter a number: 5
Factorial = 120

```
[4]: num = int(input("Enter a number: "))  
  
if num % 2 == 0:  
    print("The number is even")  
else:  
    print("The number is odd")
```

Enter a number: 8
The number is even



```
[6]: num = int(input("Enter a number: "))  
     if num>0:  
         print("The number is positive")
```

Enter a number: 2
The number is positive

```
[7]: ch = input("enter a character: ")  
     if ch in ('a' , 'e' , 'i' , 'o' , 'u' , 'A' , 'E' , 'I' , 'O' , 'U'):  
         print("Its is a vowel")
```

enter a character: A
Its is a vowel

```
[9]:  
  
number = int(input("Enter a number: "))  
  
if number % 5 == 0:  
    print(f"{number} is divisible by 5")  
else:  
    print(f"{number} is not divisible by 5")
```

Enter a number: 10
10 is divisible by 5

[10]:

```
num = int(input("Enter a number: "))

if num % 2 == 0:
    print(f"{num} is even")
else:
    print(f"{num} is odd")
```

Enter a number: 5
5 is odd

[13]:

```
a = 10
b = 20

if a > b:
    print("Largest number is:", a)
else:
    print("Largest number is:", b)
```

Largest number is: 20

[17]:

```
def is_leap_year(year):
    if (year % 400 == 0) or (year % 4 == 0 and year % 100 != 0):
        return True
    else:
        return False

# Example usage
year = 2024
print(f"{year} is a leap year: {is_leap_year(year)}")
```

2024 is a leap year: True

•[18]: *# Python program to calculate grade based on marks*

```
marks = int(input("Enter your marks: "))
```

```
if marks >= 90:
```

```
    grade = "A+"
```

```
elif marks >= 80:
```

```
    grade = "A"
```

```
elif marks >= 70:
```

```
    grade = "B"
```

```
elif marks >= 60:
```

```
    grade = "C"
```

```
elif marks >= 50:
```

```
    grade = "D"
```

```
else:
```

```
    grade = "F"
```

```
print("Your grade is:", grade)
```

Enter your marks: 95

Your grade is: A+

[19]: *# Python program to classify a number*

```
num = float(input("Enter a number: "))
```

```
if num > 0:
```

```
    print(num, "is Positive")
```

```
elif num < 0:
```

```
    print(num, "is Negative")
```

```
else:
```

```
    print(num, "is Zero")
```

Enter a number: 0

0.0 is Zero

[22]: *# Python program to find the largest of three numbers using if-elif-else*

```
a = float(input("Enter first number: "))
b = float(input("Enter second number: "))
c = float(input("Enter third number: "))

if (a >= b) and (a >= c):
    print("The largest number is:", a)
elif (b >= a) and (b >= c):
    print("The largest number is:", b)
else:
    print("The largest number is:", c)
```

Enter first number: 2
Enter second number: 4
Enter third number: 6
The largest number is: 6.0

[23]: *# Python program to print numbers from 1 to 10*

```
for i in range(1, 11):
    print(i)
```

1
2
3
4
5
6
7
8
9
10

[24]: *# Python program to print multiplication table of a given number*

```
num = int(input("Enter a number: "))

print(f"Multiplication Table of {num}")
for i in range(1, 11):
    print(f"{num} x {i} = {num * i}")
```

Enter a number: 5

Multiplication Table of 5

5 x 1 = 5

5 x 2 = 10

5 x 3 = 15

5 x 4 = 20

5 x 5 = 25

5 x 6 = 30

5 x 7 = 35

5 x 8 = 40

5 x 9 = 45

5 x 10 = 50

[25]: *# Python program to calculate factorial of a number*

```
num = int(input("Enter a number: "))

factorial = 1

if num < 0:
    print("Factorial does not exist for negative numbers")
elif num == 0:
    print("The factorial of 0 is 1")
else:
    for i in range(1, num + 1):
        factorial *= i
    print(f"The factorial of {num} is {factorial}")
```

Enter a number: 3
The factorial of 3 is 6

[26]: *# Python program to calculate sum of first n natural numbers using while loop*

```
n = int(input("Enter a number: "))

sum = 0
i = 1

while i <= n:
    sum += i
    i += 1

print(f"The sum of the first {n} natural numbers is: {sum}")
```

Enter a number: 5
The sum of the first 5 natural numbers is: 15


```
[30]: # Python program to reverse a number using while loop

num = int(input("Enter a number: "))
reverse = 0

while num > 0:
    digit = num % 10      # extract last digit
    reverse = reverse * 10 + digit # build reversed number
    num = num // 10      # remove last digit

print("Reversed number is:", reverse)
```

Enter a number: 35
Reversed number is: 53

```
[34]: # Python program to count digits in a number using while loop

num = int(input("Enter a number: "))
count = 0

# Handle zero separately
if num == 0:
    count = 1
else:
    # Work with absolute value to handle negatives
    num = abs(num)
    while num > 0:
        num = num // 10 # remove last digit
        count += 1      # increment digit count

print("Number of digits:", count)
```

Enter a number: 12345
Number of digits: 5

[35]: *# Python program to stop printing numbers when a specific number is found*

```
stop_num = int(input("Enter the number at which to stop: "))
```

```
for i in range(1, 11):
```

```
    if i == stop_num:
```

```
        print(f"Stopping at {stop_num}")
```

```
        break
```

```
    print(i)
```

Enter the number at which to stop: 5

1

2

3

4

Stopping at 5

```
[36]: # Python program to search for an element in a list and stop when found
```

```
numbers = [10, 20, 30, 40, 50, 60]
search = int(input("Enter the number to search: "))

for num in numbers:
    if num == search:
        print(f"Element {search} found in the list!")
        break
    print(f"Checked {num}, not a match.")
else:
    # This else runs only if the loop completes without break
    print(f"Element {search} not found in the list.")
```

```
Enter the number to search: 5
Checked 10, not a match.
Checked 20, not a match.
Checked 30, not a match.
Checked 40, not a match.
Checked 50, not a match.
Checked 60, not a match.
Element 5 not found in the list.
```

[37]: *# Python program to print numbers from 1 to 10 except multiples of 3*

```
for i in range(1, 11):  
    if i % 3 == 0:  
        continue # skip multiples of 3  
    print(i)
```

1
2
4
5
7
8
10

[39]: *# Python program to skip printing vowels in a string*



```
text = input("Enter a string: ")  
  
for ch in text:  
    if ch.lower() in ['a', 'e', 'i', 'o', 'u']:  
        continue # skip vowels  
    print(ch, end=" ") # print consonants only
```

Enter a string: 2 3

2 3